

Assignment 1

Advanced Computer Architecture

Date of Submission: 28/01/2026

1. What are the main types of processor architectures used in modern computers? What are the key elements and their significance in design of a processor?
2. What is a multiprocessor? Explain its architecture. Also differentiate between symmetric and asymmetric multiprocessors.
3. Explain multicomputer architecture. What is the role of Virtual Shared Memory and Shared Virtual Memory in multicomputer?
4. What is the significance of Flynn's taxonomy? Explain the Flynn's classifications with examples.
5. Compare the three multiprocessor hardware:
 - I. multiprocessor with time shared bus vs
 - II. multiprocessor with cross bar switch vs
 - III. Multiprocessors with multiport memory.